



DPUTRFDDataSingle

The **DPUTRFDDataSingle** observes a simulated vehicle specified by a UserID or a TRFID. All available data of the vehicle can be used for recording as well for connection with any other DPU.

1. Dynamic parameters:

<i>UserID</i> [1..6]	UserID of the vehicles of interest (see below)
<i>TRFID</i> [1..6]	TRFID of the vehicles of interest (see below)

2. Outputs:

<i>X</i> [1..6]	X coordinates [m]
<i>Y</i> [1..6]	Y coordinates [m]
<i>Z</i> [1..6]	Z coordinates [m]
<i>AEditX</i> [1..6]	In Area2 sections: Current position measured in the coordinate system of SILABAEEdit. In course segments always 0.
<i>AEditY</i> [1..6]	[m]
<i>yaw</i> [1..6]	Yaw angle [rad]
<i>pitch</i> [1..6]	Pitch angle [rad]
<i>roll</i> [1..6]	Roll angle in [rad]
<i>v</i> [1..6]	Speed [m/s]
<i>ax</i> [1..6]	Acceleration set for the behavior model [m/s ²]
<i>dax</i> [1..6]	Calculated acceleration [m/s ²]
<i>v_kmh</i> [1..6]	Speed of the vehicle [km/h]
<i>ax</i> [1..6]	Acceleration set for the behavior model [m/s ²]
<i>dax</i> [1..6]	Calculated acceleration [m/s ²]
<i>FloorHeight</i>	In case of a bus the height of the floor inside the vehicle
<i>LaneID</i> [1..6]	Number of the lane on which the vehicle is located
<i>Lanechange</i> [1..6]	During a lanechange this variable shows the current lateral position [m]. At the beginning of a lane change the value is 0 and then it rises until the distance between the lanes is reached.
<i>LateralDistance</i> [1..6]	Distance of the center of gravity of the vehicle to the middle of the current lane (ideal lane). Negative Distances are left of the ideal lane [m]
<i>Width</i> [1..6]	Width of the vehicle [m]
<i>WidthLeft</i> [1..6]	Width of the vehicle from center of gravity to left edge [m]
<i>WidthRight</i> [1..6]	Width of the vehicle from center of gravity to right edge [m]
<i>Length</i> [1..6]	Length of the vehicle [m]
<i>LengthFront</i> [1..6]	Length of the vehicle from center of gravity to front edge [m]
<i>LengthRear</i> [1..6]	Length of the vehicle from center of gravity to rear edge [m]
<i>Type</i> [1..6]	ODB subtype of the vehicle
<i>ID</i> [1..6]	Vehicle ID
<i>RightIndicator</i> [1..6]	Right indicator (1.0 = on; 0.0 = off)



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<code>LeftIndicator[1..6]</code>	Left indicator (1.0 = on; 0.0 = off)
<code>BrakeLight[1..6]</code>	Brake light (1.0 = on; 0.0 = off)
<code>LaneInfo[1...6]</code>	Lane information (lane number)
<code>NodeInfo[1...6]</code>	Template information about the node. In configuration file defined by NodeID.
<code>NodeInstanceInfo[1...6]</code>	Instance counter for the node
<code>ModuleInfo[1...6]</code>	Template information for the module
<code>ModuleInstanceInfo[1...6]</code>	Instance counter for the module
<code>SInfo [1..6]</code>	On Course sections: Length of the road since start of the simulation. In Area2 sections this value always is that of entering the Area2 [m]
<code>SLaneInfo[1..6]</code>	Lane position on the current lane [m]
<code>TInfo[1..6]</code>	Distance from the sides [m]

3. Dependencies on other DPUs:

DPUTRFDataSingle must run on a computer together with **DPUTRF(X)**. The DPU *Index* entry for **DPUTRFDataSingle** must be higher than the *Index* entry for **DPUTRF(X)**.

4. Operation:

DPUTRFDataSingle delivers data on vehicles that were specially highlighted by the scenario developer. For this, the developer must set the user ID of the vehicle as a parameter in the DPU, so that the relevant data is given at to the DPU outputs. Alternatively, the TRFID can be specified to identify a vehicle, for reasons of compatibility. The TRFID is a uniquely assigned ID by SILAB for every vehicle, whose value cannot however be influenced by the user. We recommend that you use the UserID parameter.

Where no matching vehicle is found, all outputs are set to 0.

5. Example:

```
DPUTRFDataSingle DataSingle
{
    Computer = { TRF };
    Index = 400;

    UserID1 = 1001;
    UserID2 = 1002;
};
```

In this example DataSingle will record all data für vehicle with UserID 1001 and UserID 1002. In the definition of vehicles these UserIDs have to be defined.

Example for defining UserIDs:

```
VehicleX Car01
{
    ...
    UserID = 1001;
```



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```
};
```

```
VehicleX Car02
```

```
{
```

```
...
```

```
    UserID = 1002;
```

```
};
```